

Polybrominated Diphenyl Ethers (PBDEs) in Breast Milk and Neuropsychological Development in Infants

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Background: There is increasing interest in the potential effects of polybrominated diphenyl ethers (PBDEs) on children's neuropsychological development, but only a few small studies have evaluated such effects. Objectives: Our goal was to examine the association between PBDE concentrations in colostrum and infant neuropsychological development and to assess the influence of other persistent organic pollutants (POPs) on such association. Methods: We measured concentrations of PBDEs and other POPs in colostrum samples of 290 women recruited in a Spanish birth cohort. We tested children for mental and psychomotor development with the Bayley Scales of Infant Development at 12–18 months of age. We analyzed the sum of the seven most common PBDE congeners (BDEs 47, 99, 100, 153, 154, 183, 209) and each congener separately. Results: Increasing Σ 7PBDEs concentrations showed an association of borderline statistical significance with decreasing mental development scores (β per log ng/g lipid = -2.25 ; 95% CI: $-4.75, 0.26$). BDE-209, the congener present in highest concentrations, appeared to be the main congener responsible for this association (β = -2.40 , 95% CI: $-4.79, -0.01$). There was little evidence for an association with psychomotor development. After adjustment for other POPs, the BDE-209 association with mental development score became slightly weaker (β = -2.10 , 95% CI: $-4.66, 0.46$). Conclusions: Our findings suggest an association between increasing PBDE concentrations in colostrum and a worse infant mental development, particularly for BDE-209, but require confirmation in larger studies. The association, if causal, may be due to unmeasured BDE-209 metabolites, including OH-PBDEs (hydroxylated PBDEs), which are more toxic, more stable, and more likely to cross the placenta and to easily reach the brain than BDE-209.